

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278475

Kind Code

A1

Publication Date

September 04, 2025

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ENHANCING INSIDER ATTACK DETECTION VIA IMAGE CLASSIFICATION WITH NON-DYNAMIC INFORMATION

Abstract

Systems, methods, devices, and apparatus are discussed for detecting potential insider attacks associated with a communication network and associated devices based on images that encode behavior activities of an insider performed in relation to a communication network, in conjunction with non-dynamic data related to the insider.

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Appl. No.: 19/202594

Filed: May 08, 2025

Related U.S. Application Data

parent US continuation 17888665 20220816 parent-grant-document US 12223038 child US 18970416

parent US continuation-in-part 18970416 20241205 PENDING child US 19202594

us-provisional-application US 63235887 20210823

Publication Classification

Int. Cl.: G06F21/55 (20130101); G06F18/24 (20230101); G06F21/31 (20130101); G06F21/62 (20130101); G06F40/157 (20200101); G06F40/205 (20200101); G06F40/242 (20200101); G06F40/279 (20200101); G06F40/284 (20200101); G06V10/56

(20220101); **G06V10/764** (20220101); **G06V10/776** (20220101); **G06V40/20** (20220101); **H04L9/40** (20220101); **H04L43/045** (20220101)

U.S. Cl.:

CPC **G06F21/552** (20130101); **G06F18/24** (20230101); **G06F21/316** (20130101); **G06F21/6218** (20130101); **G06F40/242** (20200101); **G06F40/279** (20200101); **G06F40/284** (20200101); **G06V10/56** (20220101); **G06V10/764** (20220101); **G06V10/776** (20220101); **G06V40/20** (20220101); **H04L43/045** (20130101); **H04L63/1416** (20130101); **H04L63/1425** (20130101); G06F40/157 (20200101); G06F40/205 (20200101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of part of U.S. patent application Ser. No. 18/970,416, filed Dec. 5, 2024, which is a continuation of U.S. patent application Ser. No. 17/888,665, filed Aug. 16, 2022, and issued as U.S. Pat. No. 12,223,038 on Feb. 11, 2025, which claims priority to U.S. Provisional Patent Application No. 63/235,887 entitled “Computer Vision User Entity Behavior Analytics,” filed Aug. 23, 2021, by Khanna. The entirety of the aforementioned applications are incorporated herein by reference for all purposes.

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FIELD

[0003] Embodiments discussed generally relate to systems and methods for identifying potential insider attacks on a computer network.

BACKGROUND OF THE DISCLOSURE

[0004] In general, cybersecurity efforts are focused on mitigating the risk posed to networks and organizations by malicious attacks from outside of the organization or network. However, insider threats are a significant concern in the cybersecurity landscape due to the ability of an insider to exploit insider privileges and bypass security protocols. An insider's familiarity with an organization and its systems allows the insider to mask their activities, making insider attacks difficult to detect. Accordingly, insider attacks may go unnoticed until significant damage has been done and the consequences of an insider attack may be significant.

[0005] Thus, there exists a need in the art for more advanced approaches, devices, and systems for identifying insider attacks.

SUMMARY

[0006] Various embodiments provides systems and methods for identifying potential insider attacks on a computer-based infrastructure of an organization, including but not limited to a network.

[0007] This summary provides only a general outline of some embodiments. Many other objects, features, advantages, and other embodiments will become more fully apparent from the following detailed description, the appended claims and the accompanying drawings and figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] A further understanding of the various embodiments may be realized by reference to the figures which are described in remaining portions of the specification. In the figures, similar reference numerals are used throughout several drawings to refer to similar components. In some instances, a sub-label consisting of a lower-case letter is associated with a reference numeral to denote one of multiple similar components. When reference is made to a reference numeral without specification to an existing sub-label, it is intended to refer to all such multiple similar components.

[0009] FIG. 1A-1C illustrate a network architecture including an insider attack identification and attack model training application in accordance with some embodiments;

[0010] FIG. 2A-2B are flow diagrams showing a method in accordance with some embodiments for detecting insider attacks and training an insider attack detection and classification model;

[0011] FIG. 3 is a flow diagram showing a method in accordance with some embodiments for generating one additional set of insider attack data for every actual set of insider attack data available;

[0012] FIG. 4 graphically depicts the process of the flow diagram of FIG. 3;

[0013] FIG. 5 is a flow diagram showing a method in accordance with some embodiments for generating multiple additional sets of insider attack data for every actual set of insider attack data available;

[0014] FIG. 6 graphically depicts the process of the flow diagram of FIG. 5;

[0015] FIG. 7 shows a novel GAN architecture tailored for revealing insider attacks in accordance with various embodiments;

[0016] FIG. 8 is a flow diagram showing a method of processing synthetic insider attack images in accordance with some embodiments;

[0017] FIG. 9A are example images generated using dense-convolution-dense processing in accordance with various embodiments;

[0018] FIG. 9B are example images generated without the dense-convolution-dense processing of FIG. 9A;

[0019] FIG. 10 illustrates a dual-input attack classification architecture, in accordance with some embodiments; and

[0020] FIG. 11A-11B are flow diagrams showing a method in accordance with some embodiments for detecting insider attacks and training a dual-input insider attack detection and classification model.

DETAILED DESCRIPTION

[0021] User entity and behavior analytics (UEBA) is an advanced cybersecurity approach that uses machine learning and behavioral analytics to detect compromised components of a network, as well as detecting potential malicious insiders. However, such systems generally rely on their own datasets and experience to generate rule-based implementations that take significant tuning to become effective.

[0022] Research has been performed on artificial intelligence (AI) or neural-network based systems that can classify the likelihood of an insider attack occurring within a network or organization. Computer networks process large numbers of requests from those who are authorized to operate on the computer network and from those who are not authorized. One challenge is discerning which of the requests are malicious and should not be processed. A relatively large number of unauthorized requests from outside of a network are malicious. Accordingly, it is relatively easy to train a computer model to identify such “outsider” attacks. In contrast, activities conducted by those that are authorized to access a network are a small percentage of malicious behavior. The small percentage of attacks conducted by insiders complicates the training of neural network models to detect and/or classify insider attacks, as there is only a very small amount of data related to actual insider attacks when compared with the volume of other data. Accordingly, naïve approaches to

model training may result in spurious indicators of insider attacks, which may result in a number of nonsensical false positives.

[0023] Embodiments described herein provide systems and methods for identifying potential insider attacks on a computer network and/or network connected infrastructure within an organization.

[0024] Some embodiments provide methods for detecting insider attacks that generate color behavioral images representing activities of an insider over a period, and applying an insider attack classification model to the color behavioral image to discern whether an insider attack has occurred. Where an insider attack has occurred, the color behavioral image is characterized as an actual insider attack image. This actual insider attack image may be processed to generate one or more additional synthetic insider attack images. The combination of the actual insider attack image and the one or more synthetic attack images are used to train the insider attack classification model.

[0025] Some embodiments discussed herein provide approaches for multiplying the amount of insider attack data available to train an insider attack detection model by generating one or more synthetic sets of insider attack data sets for each actual insider attack data set available. By using this additional synthetic data as representing insider attacks, the accuracy of a trained insider attack detection model can be increased. Additionally, some embodiments use a Generative Adversarial Network (GAN) to process both synthetic insider attack images and actual insider attack images to produce additional synthetic insider attack images.

[0026] Some embodiments discussed herein enhance the process of assessing and handling user behavior in a deployment setting via the use of a dual-input classifier, which considers non-dynamic information when classifying insider behavior as potentially malicious. This non-dynamic information includes, for example, an insider's role within an organization or known psychometric information about the insider. The non-dynamic data is vectorized and concatenated with image-based behavior encoding data to enhance the accuracy of insider attack detection.

[0027] Embodiments of the present disclosure include various processes, which will be described below. The processes may be performed by hardware components or may be embodied in machine-executable instructions, which may be used to cause a general-purpose or special-purpose processor programmed with the instructions to perform the steps. Alternatively, processes may be performed by a combination of hardware, software, firmware, and/or by human operators.

[0028] Embodiments of the present disclosure may be provided as a computer program product, which may include a machine-readable storage medium tangibly embodying thereon instructions, which may be used to program a computer (or other electronic devices) to perform a process. The machine-readable medium may include, but is not limited to, fixed (hard) drives, magnetic tape, floppy diskettes, optical disks, compact disc read-only memories (CD-ROMs), and magneto-optical disks, semiconductor memories, such as ROMs, PROMs, random access memories (RAMs), programmable read-only memories (PROMs), erasable PROMs (EPROMs), electrically erasable PROMs (EEPROMs), flash memory, magnetic or optical cards, or other type of media/machine-readable medium suitable for storing electronic instructions (e.g., computer programming code, such as software or firmware).

[0029] Various methods described herein may be practiced by combining one or more machine-readable storage media containing the code according to the present disclosure with appropriate standard computer hardware to execute the code contained therein. An apparatus for practicing various embodiments of the present disclosure may involve one or more computers (or one or more processors within a single computer) and storage systems containing or having network access to computer program(s) coded in accordance with various methods described herein, and the method steps of the disclosure could be accomplished by modules, routines, subroutines, or subparts of a computer program product.

[0030] While insider behavior described herein generally is described in the context of activities related to a computer network, in some embodiments, behavior may be logged and encoded for

interactions with network connected devices, such as keycard readers or other physical access mechanisms.

[0031] In the following description, numerous specific details are set forth in order to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to one skilled in the art that embodiments of the present disclosure may be practiced without some of these specific details.

Terminology

[0032] Brief definitions of terms used throughout this application are given below.

[0033] The terms “connected” or “coupled” and related terms, unless clearly stated to the contrary, are used in an operational sense and are not necessarily limited to a direct connection or coupling. Thus, for example, two devices may be coupled directly, or via one or more intermediary media or devices. As another example, devices may be coupled in such a way that information can be passed there between, while not sharing any physical connection with one another. Based on the disclosure provided herein, one of ordinary skill in the art will appreciate a variety of ways in which connection or coupling exists in accordance with the aforementioned definition.

[0034] If the specification states a component or feature “may”, “can”, “could”, or “might” be included or have a characteristic, that particular component or feature is not required to be included or have the characteristic.

[0035] As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

[0036] The phrases “in an embodiment,” “according to one embodiment,” and the like generally mean the particular feature, structure, or characteristic following the phrase is included in at least one embodiment of the present disclosure, and may be included in more than one embodiment of the present disclosure. Importantly, such phrases do not necessarily refer to the same embodiment.

[0037] As used herein, a “network appliance” or a “network device” generally refers to a device or appliance in virtual or physical form that is operable to perform one or more network functions. In some cases, a network appliance may be a database, a network server, or the like. Some network devices may be implemented as general-purpose computers or servers with appropriate software operable to perform the one or more network functions. Other network devices may also include custom hardware (e.g., one or more custom Application-Specific Integrated Circuits (ASICs)). Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a variety of network appliances that may be used in relation to different embodiments. In some cases, a network appliance may be a “network security appliance” or a network security device” that may reside within the particular network that it is protecting, or network security may be provided as a service with the network security device residing in the cloud. For example, while there are differences among network security device vendors, network security devices may be classified in three general performance categories, including entry-level, mid-range, and high-end network security devices. Each category may use different types and forms of central processing units (CPUs), network processors (NPs), and content processors (CPs). NPs may be used to accelerate traffic by offloading network traffic from the main processor. CPs may be used for security functions, such as flow-based inspection and encryption. Entry-level network security devices may include a CPU and no co-processors or a system-on-a-chip (SoC) or system-in-package (SIP) processor that combines a CPU, a CP and an NP. Mid-range network security devices may include a multi-core CPU, a separate NP Application-Specific Integrated Circuits (ASIC), and a separate CP ASIC. At the high-end, network security devices may have multiple NPs and/or multiple CPs. A network security device is typically associated with a particular network (e.g., a private enterprise network) on behalf of which it provides the one or more security functions. Non-limiting examples of security functions include authentication, next-generation firewall protection, antivirus scanning,

content filtering, data privacy protection, web filtering, network traffic inspection (e.g., secure sockets layer (SSL) or Transport Layer Security (TLS) inspection), intrusion prevention, intrusion detection, denial of service attack (DoS) detection and mitigation, encryption (e.g., Internet Protocol Secure (IPSec), TLS, SSL), application control, Voice over Internet Protocol (VoIP) support, Virtual Private Networking (VPN), data leak prevention (DLP), antispam, antispyware, logging, reputation-based protections, event correlation, network access control, vulnerability management, and the like. Such security functions may be deployed individually as part of a point solution or in various combinations in the form of a unified threat management (UTM) solution. Non-limiting examples of network security appliances/devices include network gateways, VPN appliances/gateways, UTM appliances (e.g., the FORTIGATE family of network security appliances), messaging security appliances (e.g., FORTIMAIL family of messaging security appliances), database security and/or compliance appliances (e.g., FORTIDB database security and compliance appliance), web application firewall appliances (e.g., FORTIWEB family of web application firewall appliances), application acceleration appliances, server load balancing appliances (e.g., FORTIBALANCER family of application delivery controllers), network access control appliances (e.g., FORTINAC family of network access control appliances), vulnerability management appliances (e.g., FORTISCAN family of vulnerability management appliances), configuration, provisioning, update and/or management appliances (e.g., FORTIMANAGER family of management appliances), logging, analyzing and/or reporting appliances (e.g., FORTIANALYZER family of network security reporting appliances), bypass appliances (e.g., FORTIBRIDGE family of bypass appliances), Domain Name Server (DNS) appliances (e.g., FORTIDNS family of DNS appliances), wireless security appliances (e.g., FORTIWIFI family of wireless security gateways), virtual or physical sandboxing appliances (e.g., FORTISANDBOX family of security appliances), and DoS attack detection appliances (e.g., the FORTIDDOS family of DOS attack detection and mitigation appliances).

[0038] The phrase “processing resource” is used in its broadest sense to mean one or more processors capable of executing instructions. Such processors may be distributed within a network environment or may be co-located within a single network appliance. Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a variety of processing resources that may be used in relation to different embodiments.

[0039] The phrase “insider attack” is used in its broadest sense to mean any attack against or launched from a communication network where the perpetrator of the attack is a trusted insider. As one example, a trusted insider may be someone who has been granted permission to access the communication network and has accessed the communication network using such permission. This is in contrast to an outsider who has not been granted permission to access the communication network, but may have obtained access through illicit means. In some cases, an insider attack is made by a trusted insider who has accessed the communication network from within a trusted perimeter. Such a trusted perimeter may be, but is not limited to, within a building supported by the communication network using, for example, a computer assigned to the trusted insider that is connected to the communication network physically within the building.

[0040] Example embodiments will now be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments are shown. This disclosure may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. It will be appreciated by those of ordinary skill in the art that the diagrams, schematics, illustrations, and the like represent conceptual views of processes illustrating systems and methods embodying various aspects of the present disclosure. The functions of the various elements shown in the figures may be provided through the use of dedicated hardware as well as hardware capable of executing associated software and their functions may be carried out through the operation of program logic, through dedicated logic, through the interaction of program control and dedicated logic.

[0041] Various embodiments provide methods for detecting insider attacks. The methods include: logging, by a processing resource, activities of an insider performed in relation to a communication network to yield logged activities; extracting, by the processing resource, a set of defined behavioral features from the logged activities; forming, by the processing resource, a grayscale behavioral image corresponding to the set of defined behavioral features; combining, by the processing resource, the grayscale behavioral image with a first grayscale attack context image and a second grayscale attack context image to yield a color behavioral image; applying, by the processing resource, an insider attack classification model to the color behavioral image, wherein the color behavioral image indicates an actual insider attack; and storing, by the processing resource, the color behavioral image as an actual insider attack image.

[0042] In some instances of the aforementioned embodiments, the methods further include generating, by the processing resource, at least one synthetic insider attack image using the actual insider attack image. In some such instances, the methods further include: training, by the processing resource, the insider attack classification model using a combination of images including at least the actual insider attack image and the at least one synthetic insider attack image. In some instances of the aforementioned embodiments where the at least one synthetic insider attack image is a first insider attack image, the methods further include applying, by the processing resource, a generative adversarial network to the actual insider attack image to generate at least a second synthetic insider attack image. In some such instances, the methods further include training, by the processing resource, the insider attack classification model using a combination of images including at least the actual insider attack image, the first synthetic insider attack image, and the second insider attack image. In various instances of the aforementioned embodiments where the at least one synthetic insider attack image is a first insider attack image, the methods further include applying, by the processing resource, a generative adversarial network to the first synthetic insider attack image to generate at least a second synthetic insider attack image. In some such instances, the methods further include training, by the processing resource, the insider attack classification model using a combination of images including at least the actual insider attack image, the first synthetic insider attack image, and the second insider attack image.

[0043] In various instances of the aforementioned embodiments, the methods further include receiving, by the processing resource, a corroboration from an expert that the color behavioral image indicates the actual insider attack. In such instances, the storing the color behavioral image as an actual insider attack image is based upon a combination of the indication of the actual insider attack from the insider attack classification model and the corroboration from the expert. In some instances of the aforementioned embodiments combining the grayscale behavioral image with a first grayscale attack context image and a second grayscale attack context image to yield a color behavioral image includes: assigning the values of the grayscale behavioral image as one of a red component, a green component, or a blue component (e.g., a red component); assigning the values of the first grayscale attack context image as another of the red component, the green component, or the blue component (e.g., a green component); assigning the values of the second grayscale attack context image as a remaining one the red component, the green component, or the blue component (e.g., a blue component). In such instances, the color behavioral image includes a number of pixel positions each with red, green, and blue components corresponding to the same pixel position in each of the grayscale behavioral image, the first grayscale attack context image, and the second grayscale attack context image.

[0044] Other embodiments provide computer readable media that have stored therein instructions, which when executed by a processor, causes the processor to perform the following method: logging activities of an insider performed in relation to a communication network to yield logged activities; extracting a set of defined behavioral features from the logged activities; forming a grayscale behavioral image corresponding to the set of defined behavioral features; combining the grayscale behavioral image with a first grayscale attack context image and a second grayscale

attack context image to yield a color behavioral image; applying an insider attack classification model to the color behavioral image, wherein the color behavioral image indicates an actual insider attack; and storing the color behavioral image as an actual insider attack image.

[0045] Yet other embodiments provide systems that include a processing resource and a computer readable medium. The computer readable medium has stored therein instructions, which when executed by the processing resource, causes the processing resource to: log activities of an insider performed in relation to a communication network to yield logged activities; extract a set of defined behavioral features from the logged activities; form a grayscale behavioral image corresponding to the set of defined behavioral features; combine the grayscale behavioral image with a first grayscale attack context image and a second grayscale attack context image to yield a color behavioral image; apply an insider attack classification model to the color behavioral image, wherein the color behavioral image indicates an actual insider attack; and store the color behavioral image as an actual insider attack image.

[0046] Various embodiments provide methods for increasing insider attack data. Such methods include: accessing an actual insider attack image; disaggregating the actual insider attack image into a first component image corresponding to an actual insider attack, a second component image corresponding to a first context of the actual insider attack, and a third component image corresponding to a first context of the actual insider attack; replacing the second component image with an alternate component image; and aggregating the first component image with at least the alternate component image to yield a synthetic insider attack image.

[0047] In some instances of the aforementioned embodiments, the alternate component image corresponds to a third context of the actual insider attack. In some such instances, the first context represents a different time period than the third context. In various of such instances, the third context is an average of context data over multiple time periods.

[0048] In various instances of the aforementioned embodiments where the alternate image is a first alternate image, the methods further include: replacing the third component image with a second alternate component image. In such instances, aggregating the first component image includes aggregating the first component image with the first alternate image and the second alternate image to yield the synthetic insider attack image. In some cases the first component image is a first color component of the actual insider attack image, the second component image is a second color component of the actual insider attack image, and the third component image is a third color component of the actual insider attack images. In one or more cases, the first color component is one of a red component, a green component, or a blue component; the second color component is another one of the red component, the green component, or the blue component; and the third color component is the remaining one of the red component, the green component, or the blue component.

[0049] In various of such instances, the first alternate component image is the third component image, and the second alternate component image is the second component image. In some cases, the first alternate component image corresponds to a third context of the actual insider attack, and the second alternate component image corresponds to a fourth context of the actual insider attack. In some cases, the first context represents a different time period than the third context, and wherein the second context represents a different time period than the fourth context. In more or more cases, the third context is an average of context data over multiple time periods.

[0050] Other embodiments provide computer readable media having stored therein instructions, which when executed by a processor, causes the processor to perform the following method: accessing an actual insider attack image; disaggregating the actual insider attack image into a first component image corresponding to an actual insider attack, a second component image corresponding to a first context of the actual insider attack, and a third component image corresponding to a first context of the actual insider attack; replacing the second component image with an alternate component image; and aggregating the first component image with at least the

alternate component image to yield a synthetic insider attack image.

[0051] Yet other embodiments provide systems for augmenting insider attack data that include a processing resource and a computer readable medium. The computer readable medium has stored therein instructions, which when executed by the processing resource, causes the processing resource to: access an actual insider attack image; disaggregate the actual insider attack image into a first component image corresponding to an actual insider attack, a second component image corresponding to a first context of the actual insider attack, and a third component image corresponding to a first context of the actual insider attack; replace the second component image with an alternate component image; and aggregate the first component image with at least the alternate component image to yield a synthetic insider attack image.

[0052] Some embodiments provide methods for modeling insider attacks. Such methods include: generating an image for each of multiple attack vectors to generate multiple real images, where each of the multiple attack vectors include data elements from network traffic in an insider attack; creating multiple noise modified vectors by modifying at least one element of each of the multiple attack vectors; by the processing resource, an image for each of multiple noise modified vectors to yield multiple generated images; classifying each of the real images and generated images as corresponding to either an attack vector or a noise modified vector; calculating an accuracy of the classifying; and identifying an insider attack model as trained based at least in part on the accuracy of the classifying being less than or equal to a threshold value.

[0053] In some instances of the aforementioned embodiments, the threshold value is between forty-five (45) percent accurate and fifty-five (55) percent accurate. In various instances of the aforementioned embodiments, the threshold value is between than forty-nine (49) percent accurate and fifty-one (51) percent accurate.

[0054] In various instances of the aforementioned embodiments, generating an image of an attack vector includes: applying a leaky rectified linear unit fully connected layer stage to the attack vector to yield a first output; and applying a leaky rectified linear unit two dimension convolution stage to a second output derived from the first output to yield a third output. In some such instances where the leaky rectified linear unit fully connected layer stage is a first leaky rectified linear unit fully connected layer stage and the leaky rectified linear unit two dimension convolution stage is a first leaky rectified linear unit two dimension convolution stage, generating the image of the attack vector further includes: applying a second leaky rectified linear unit fully connected layer stage to a fourth output derived from the first output to yield a fifth output, wherein the second output is derived from the fifth output; applying a second leaky rectified linear unit two dimension convolution stage to a sixth output derived from the third output to yield a seventh output; and applying a third leaky rectified linear unit two dimension convolution stage to an eighth output derived from the seventh output.

[0055] In some instances of the aforementioned embodiments, generating an image of a noise modified vector includes: applying the leaky rectified linear unit fully connected layer stage to the attack vector to yield a fourth output; and applying the leaky rectified linear unit two dimension convolution stage to a fifth output derived from the fourth output to yield a sixth output.

[0056] In various instances of the aforementioned embodiments, classifying an image includes: flattening a received image to yield a flattened image; applying a leaky rectified linear unit fully connected layer stage to the flattened image to yield a first output; and applying a leaky rectified linear unit two dimension convolution stage to a second output derived from the first output to yield a third output. In some such instances where the leaky rectified linear unit fully connected layer stage is a first leaky rectified linear unit fully connected layer stage and the leaky rectified linear unit two dimension convolution stage is a first leaky rectified linear unit two dimension convolution stage, generating the image of the attack vector further includes: applying a second leaky rectified linear unit two dimension convolution stage to the third output to yield a fifth output; applying a third leaky rectified linear unit two dimension convolution stage to the fifth

output to yield a sixth output; and applying a second leaky rectified linear unit fully connected layer stage to a seventh output derived from the sixth output.

[0057] Other embodiments provide systems for training an insider attack classification model. Such systems include a processing resource and a non-transitory computer-readable medium. The non-transitory computer-readable medium has stored therein instructions that when executed by the processing resource cause the processing resource to perform the method including: generating an image for each of multiple attack vectors to generate multiple real images, where each of the multiple attack vectors include data elements from network traffic in an insider attack; creating multiple noise modified vectors by modifying at least one element of each of the multiple attack vectors; generating an image for each of multiple noise modified vectors to yield multiple generated images; classifying each of the real images and generated images as corresponding to either an attack vector or a noise modified vector; calculating an accuracy of the classifying; and identifying an insider attack model as trained based at least in part on the accuracy of the classifying being less than or equal to a threshold value.

[0058] Yet other embodiments provide non-transitory computer-readable storage media embodying a set of instructions, which when executed by a processing resource, causes the processing resource to perform a method including: generating an image for each of multiple attack vectors to generate multiple real images, where each of the multiple attack vectors include data elements from network traffic in an insider attack; creating multiple noise modified vectors by modifying at least one element of each of the multiple attack vectors; generating an image for each of multiple noise modified vectors to yield multiple generated images; classifying each of the real images and generated images as corresponding to either an attack vector or a noise modified vector; calculating an accuracy of the classifying; and identifying an insider attack model as trained based at least in part on the accuracy of the classifying being less than or equal to a threshold value.

[0059] Other embodiments provide systems, methods, and non-transitory computer-readable storage media employing a set of instructions to detect insider attacks based on encoded insider behavior and non-dynamic information (e.g., job role, psychometric data, etc.) associated with the insider. A multi-input (e.g., dual input) classifier may be trained that enhances image-based classification techniques with the addition of non-dynamic information to increase the accuracy of the classification output.

[0060] For example, a method for detecting insider attacks, or behavior that is indicative of an insider attack, can comprise logging, by a processing resource, activities of an insider performed in relation to a communication network to yield logged activities; extracting, by the processing resource, a set of defined behavioral features from the logged activities; forming, by the processing resource, color behavioral image based on a grayscale behavioral image corresponding to the set of defined behavioral features and one or more grayscale behavioral context images; and applying, by the processing resource, a multiple-input insider attack classification model to the color behavioral image and non-dynamic data associated with the insider, the non-dynamic data including an organizational role of the insider.

[0061] Additionally a non-transitory computer-readable medium having instructions stored therein may be provided, where the instructions when executed by a processing resource, cause the processing resource to perform operations including logging activities associated with an electronic account of an insider within an organization, the activities performed in relation to a communication network to yield logged activities; extracting a set of defined behavioral features from the logged activities; forming color behavioral image based on a grayscale behavioral image corresponding to the set of defined behavioral features and one or more grayscale behavioral context images; and applying a multiple-input insider attack classification model to the color behavioral image and non-dynamic data associated with the insider, the non-dynamic data including an organizational role of the insider.

[0062] In addition to the above, an apparatus may be provided that is configurable to classify user

behavior, where the apparatus comprises a memory device configured to store instructions; and a processing resource configurable to execute the instructions. The instructions configure the processing resource to generate a color behavioral image based on a grayscale behavioral image corresponding to a set of defined behavioral features logged for an electronic account associated with an insider of an organization and one or more grayscale behavioral context images; process the color behavioral image via a first neural network model; concatenate output of the first neural network model with non-dynamic data associated with the insider to generate concatenated data; process the concatenated data via a second neural network model to generate a classification score; and determine whether the classification score indicates an insider attack based on a comparison with an insider attack classification threshold.

[0063] Turning to FIG. 1A, network architecture **100** is shown in accordance with some embodiments. In the context of network architecture **100**, a network security appliance **103** controls access to network elements within a local network **101** derived from outsider endpoint devices **107a-107b** (e.g., outsider endpoint device A **107a** and outsider endpoint device B **107b**). Further, network security appliance **103** monitors network traffic derived from insider endpoint devices **105** (e.g., insider endpoint device A **105a**, insider endpoint device B **105b**, and insider endpoint device C **105c**). An insider endpoint device **105** is an endpoint device that is known to and has been authorized to access local network **101**. In contrast, an outsider endpoint device has not been authorized to access elements in local network **101**.

[0064] Insider endpoint devices **105** may be used, for example, to access a shared file system **198** supported by a server **102** or to access a shared file system **199** supported by a server **108**. Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a variety of network services and/or devices that may be accessed by insider endpoint devices **105**, but not accessed by outsider endpoint devices.

[0065] Local network **101** may be any type of communication network known in the art. Those skilled in the art will appreciate that, local network **101** can be a wireless network, a wired network, or a combination thereof that can be implemented as one of the various types of networks, such as an Intranet, a Local Area Network (LAN), a Wide Area Network (WAN), an Internet, and the like. Further, local network **101** can either be a dedicated network or a shared network. The shared network represents an association of the different types of networks that use a variety of protocols, for example, Hypertext Transfer Protocol (HTTP), Transmission Control Protocol/Internet Protocol (TCP/IP), Wireless Application Protocol (WAP), and the like.

[0066] Network security appliance **103** is communicably coupled to a computer readable medium that includes an insider attack identification and attack model training application **104**. The application includes instructions executable by network security appliance **103** to generate synthetic insider attack data sets based upon actual insider attack data sets. These synthetic insider attack data sets are used in addition to the actual insider attack data sets to train an insider attack detection model. This insider attack detection model may be deployed in relation to network security appliance **103** to monitor insider attacks on local network **101**.

[0067] Turning to FIG. 1B, an example implementation of network security appliance **103** executing insider attack identification and attack model training application **104** is shown in accordance with some embodiments. Among other things, network security appliance **103** includes: an insider behavior monitoring and insider attack detection model **111**, an insider attack data multiplication module **127**, an insider attack image processing module **135**, and an insider attack detection and classification model training module **148**.

[0068] Insider behavior monitoring and insider attack detection model **111** includes: an insider monitoring module **112**, an insider behavior extraction module **114**, a grayscale insider behavior image formation module **116**, a color insider attack generation module **118**, an insider attack detection and confirmation module **120**, and an insider attack image storage module **122**.

[0069] Insider monitoring module **112** is configured to detect interaction of an insider with a

communication network and to log the insider's activities. Such activities may include, but are not limited to, file systems accessed, files accessed, computers used, applications used, time the activities occur (e.g., timestamp), use of external storage media such as thumb drives, or the like. Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a large variety of activities that may be logged in relation to the trusted insider in relation to different embodiments.

[0070] Insider behavior extraction module **114** is configurable to periodically extract behavioral features of the insider from the logged insider activity. The features that are extracted are user programmable and are typically chosen as those expected to be seen when problematic behavior is ongoing. Such features may include, but are not limited to, time of day, files accessed, file systems accessed, files accessed, computers used, applications used, use of external storage media such as thumb drives, or the like.

[0071] Grayscale insider behavior image formation module **116** is configured to form the extracted behavioral features into a grayscale behavioral image. This may be done, for example, by taking a behavior vector including each of the respective features in a defined location in the behavior vector and form it into a two dimensional image (e.g., a 32×32 image where the vector includes 1024 or fewer features). In some embodiments, the grayscale insider behavior image formation module **116** is configured to generate a feature array based on the extracted behavioral features and applying a sparse auto encoder to the feature array. Output of the sparse autoencoder may then be transformed into output values into a range of 0-255, or some other range that is suitable for encoding into a grayscale image.

[0072] Color insider behavior image formation module **118** is configurable to access a first grayscale insider behavioral image and a second grayscale insider behavioral image, and to combine the grayscale behavioral images to yield a color behavioral image. The first grayscale insider behavioral image and the second grayscale insider behavioral image are generated from data of prior time periods for the same trusted individual. As an example, in some embodiments, the first grayscale insider behavioral image and a second grayscale insider behavioral image are the preceding days' image and the image from two days prior, respectively, for the same insider being monitored. As another example, the first grayscale insider behavioral image and a second grayscale insider behavioral image are the preceding days' data and an average of the data for the preceding month, respectively, for the same insider being monitored. The first grayscale insider behavioral image and a second grayscale insider behavioral image may be considered context images provide an encoding of the behavior of the user over some previous period of time. Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a variety of context images that may be used in relation to different embodiments. The context images may be combined with a grayscale behavioral image that is generated for a given day of user behavior to be analyzed to yield a color behavioral image. Combining the aforementioned grayscale images to yield the color behavioral image may include assigning a respective component color (e.g., red, green, or blue) to each of a grayscale behavioral image, a first grayscale behavioral context image, and a second grayscale behavioral context image. As a particular example, the values of the grayscale behavioral image are assigned as red values, the values of the first grayscale behavioral context image are assigned as green values, and the second grayscale behavioral context image are assigned as blue values. When combined, each location of the resulting image has a complemented of red, green, and blue values which render the color image.

[0073] Insider attack detection and confirmation module **120** is configured to apply an insider attack classification model to the color behavioral image. This application results in an indication of whether the color behavioral image indicates an insider attack. This model is trained using actual insider attack images and synthetic insider attack images as discussed below, and the more images available for training generally result in an increased accuracy of the model.

[0074] Insider attack image storage module **122** is configurable to store a color behavioral image

that is classified as an actual insider attack image. In some cases, the classification is based only on the output of the insider attack classification model. In other cases, the classification is based upon both the output of the insider attack classification model and an input from an expert reviewing the data corresponding to the color behavioral image.

[0075] Insider attack data multiplication module **127** is configurable to generate a number of synthetic insider attack images for each actual insider attack image that is available. Insider attack data multiplication module **127** includes an insider attack image access module **128**, an insider attack image disaggregation module **130**, an insider attack context modification module and synthetic insider attack generation module **132**, and a synthetic attack image storage module **134**.

[0076] Insider attack image access module **128** is configurable to select and access insider attack images from a storage medium, and select and access alternate context images where used. Insider attack image disaggregation module **130** is configurable to disaggregate selected insider attack images into component parts. The component parts include a first component part corresponding to an actual insider attack, and a second and third component parts corresponding to context for the actual insider attack.

[0077] Insider attack context modification module and synthetic insider attack generation module **132** is configurable to replace one or both of the second and third component parts with alternate contexts. The alternate contexts pertain to the same individual associated with the selected insider attack image. In some cases, the alternate context(s) is/are replacement contexts not previously associated with the insider attack image. In other cases, the alternate contexts are simply the second and third component parts, and the replacement includes swapping the second component part with the third component part. In addition, insider attack context modification module and synthetic insider attack generation module **132** is configurable to create a synthetic insider attack image by combining the first component part, the second component part, and the third component part after component part replacement. Insider attack image storage module **142** is configurable to store the generated synthetic insider attack image with other insider attack images including, but not limited to, the actual insider attack image from which the synthetic insider attack image was generated.

[0078] Insider attack image processing module **135** is configurable to process insider attack images (both actual insider attack images and synthetic insider attack images) to reduce effects of generating multiple synthetic insider attack images based upon fewer actual insider attack images such as, for example, overfitting. Insider attack image processing module **135** includes an insider attack image access module **136**, a Wasserstein GAN with gradient penalty (W GAN-GP) enhanced with dense-convolution-dense processes module **138**, a critic module **140**, and an insider attack image storage module **142**.

[0079] Insider attack image access module **136** is configurable to access insider attack images from a storage medium, and reduce the insider attack images to vectors representative of a corresponding insider attack (whether synthetic or actual). These vectors are provided to WGAN-GP enhanced with dense-convolution-dense processes module **138**.

[0080] WGAN-GP enhanced with dense-convolution-dense processes module **138** is configurable to generate images based upon the vectors received from insider attack image access module **136**. In addition, WGAN-GP enhanced with dense-convolution-dense processes module **138** introduces noise to the received insider attack vectors by making small modifications to different vector values and generates images for each of these noise modified vectors. Such an approach results in a significant multiplication of images for the actual insider attack vectors and for surrounding noise modified vectors. Such an approach yields potentially thousands of slightly different images for each of the actual insider attack vectors.

[0081] Critic module **140** is configurable to accept a generated image from WGAN-GP enhanced with dense-convolution-dense processes module **138** and classify the received image as a real image or a generated image. In effect, critic module **140** seeks to determine whether an image received from WGAN-GP enhanced with dense-convolution-dense processes module **138** was

created using a vector from an actual insider attack or a noise modified vector. Where the noise modification was to vector elements that are not indicative of an insider attack, critic module **140** is able to classify the received image as not corresponding to an insider attack or a generated image. In contrast, where the noise modification was to vector elements that are indicative of an insider attack, critic module **140** is less able to correctly classify the received image as a generated image. This classification information is fed back to WGAN-GP enhanced with dense-convolution-dense processes module **138**.

[0082] In operation, WGAN-GP enhanced with dense-convolution-dense processes module **138** begins by adding noise to random elements in a known insider attack vectors to generate a large number of images. These images are provided to critic module **140** that operates to classify the received images as either real images (i.e., generated from an actual insider attack) or generated images (i.e., generated from a noise modified vector). This classification information is fed back to WGAN-GP enhanced with dense-convolution-dense processes module **138**. Where a noise modified vector is incorrectly classified by critic module **140** as a real image, WGAN-GP enhanced with dense-convolution-dense processes module **138** learns that modification around the vector element modified in the particular noise modified vector confuses critic module **140**. In contrast, where a noise modified vector is correctly classified by critic module **140** as a generated image, WGAN-GP enhanced with dense-convolution-dense processes module **138** learns that modification around the vector element modified in the particular noise modified vector does not confuse critic module **140**. Insider attack image storage module **142** is configurable to store images of all vectors (whether real or generated) that are identified by critic module **140** as indicative of an insider attack.

[0083] Insider attack detection and classification model training module **148** is configurable to access a combination of actual insider attack images and synthetic insider attack images to train and update training of the insider attack detection and classification model used to detect insider attacks in a communication network. Such training may be done in accordance with any known machine learning approach known. As more insider attack images become available, the training of the insider attack detection and classification model results in more accurate understanding of what an insider attack looks like, and thus provides more accuracy in detecting and classifying insider attacks.

[0084] Turning to FIG. **1C**, an example computer system **160** is shown in which or with which embodiments of the present disclosure may be utilized. As shown in FIG. **1C**, computer system **160** includes an external storage device **170**, a bus **172**, a main memory **174**, a read-only memory **176**, a mass storage device **178**, communication port(s) (e.g., communication port **180**), and one or more processing resources (e.g., processing resource **182**), which may be or include one or more processors or processor cores. The processing resource **182** may be a general-purpose processing resource or a more specialized processing resource that includes an instruction set architecture that provides both general-purpose and special-purpose instructions. In one embodiment, computer system **160** may represent some portion of any of network security appliance **103**, insider endpoint device **105**, and/or outsider endpoint device.

[0085] Those skilled in the art will appreciate that computer system **160** may include more than one processing resource **182** and communication port **180**. Non-limiting examples of processing resources include, but are not limited to, Intel Quad-Core, Intel i3, Intel i5, Intel i7, Apple M1, AMD Ryzen, or AMD® Opteron® or Athlon MP® processor(s), Motorola® lines of processors, FortiSOC™ system on chip processors or other future processors. The processing resource **182** may include various modules associated with embodiments of the present disclosure.

[0086] Communication port **180** can be any of an RS-232 port for use with a modem-based dialup connection, a 10/100 Ethernet port, a Gigabit, 10 Gigabit, 25G, 40G, and 100G port using copper or fiber, a serial port, a parallel port, or other existing or future ports. Communication port **180** may be chosen depending on a network, such as a Local Area Network (LAN), Wide Area Network

(WAN), or any network to which the computer system connects.

[0087] Main memory **174** can be Random Access Memory (RAM), or any other dynamic storage device commonly known in the art. Read-only memory **176** can be any static storage device(s) e.g., but not limited to, a Programmable Read Only Memory (PROM) chips for storing static information e.g., start-up or BIOS instructions for the processing resource.

[0088] The mass storage device **178** may be any current or future mass storage solution, which can be used to store information and/or instructions. Non-limiting examples of mass storage solutions include Parallel Advanced Technology Attachment (PATA) or Serial Advanced Technology Attachment (SATA) hard disk drives or solid-state drives (internal or external, e.g., having Universal Serial Bus (USB) and/or Firewire interfaces), e.g. those available from Seagate (e.g., the Seagate Barracuda **7200** family) or Hitachi (e.g., the Hitachi Deskstar **7K 1300**), one or more optical discs, Redundant Array of Independent Disks (RAID) storage, e.g. an array of disks (e.g., SATA arrays), available from various vendors including Dot Hill Systems Corp., LaCie, Nexsan Technologies, Inc. and Enhance Technology, Inc.

[0089] Bus **172** communicatively couples processing resource(s) with the other memory, storage and communication blocks. Bus **172** can be, e.g., a Peripheral Component Interconnect (PCI)/PCI Extended (PCI-X) bus, Small Computer System Interface (SCSI), USB or the like, for connecting expansion cards, drives and other subsystems as well as other buses, such as front side bus (FSB), which connects processing resources to software systems.

[0090] Optionally, operator and administrative interfaces, e.g., a display, keyboard, and a cursor control device, may also be coupled to bus **172** to support direct operator interaction with the computer system. Other operator and administrative interfaces can be provided through network connections connected through communication port **180**. External storage device **190** can be any kind of external hard-drives, floppy drives, IOMEGA® Zip Drives, Compact Disc-Read Only Memory (CD-ROM), Compact Disc-Rewritable (CD-RW), Digital Video Disk-Read Only Memory (DVD-ROM). Components described above are meant only to show various possibilities. In no way should the aforementioned example computer systems limit the scope of the present disclosure.

[0091] Turning to FIG. **2A**, a flow diagram **200** shows a method in accordance with some embodiments for detecting insider attacks and training an insider attack detection and classification model. Following flow diagram **200**, an indication of a network access granted to an insider is received (block **202**). This may include any indication that a trusted insider is using the communication network. As an example, an indication may be received that the trusted insider entered the building supported by the communication network. As another example, an indication may be received that a computer assigned to the trusted insider and connected to the communication network is being used. As yet another example, an indication may be received that the trusted insider has been authenticated to the communication network. Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a variety of indications that may be received that indicate a start of activities by a trusted insider.

[0092] All of the trusted insider's activities are logged (block **204**). Such activities may include, but are not limited to, file systems accessed, files accessed, computers used, applications used, time the activities occur, use of external storage media such as thumb drives, or the like. Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a large variety of activities that may be logged in relation to the trusted insider in relation to different embodiments. It is determined whether an attack review period has completed (block **206**). Such an attack review period may be any period of time over which an insider's activities are logged. For example, in some embodiments, the attack review period is a twenty-four (24) hour period that begins at midnight one day and ends at midnight the following day. Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a variety of attack review periods that may be used in relation to different embodiments.

[0093] Where the attack review period has completed (block **206**), behavioral features are extracted from the monitored insider activity (block **208**). The features that are extracted are user programmable and are typically chosen as those expected to be seen when problematic behavior is ongoing. Such features may include, but are not limited to, time of day, files accessed, file systems accessed, files accessed, computers used, applications used, use of external storage media such as thumb drives, or the like. Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a large variety of features that may be extracted from monitored insider activity in accordance with different embodiments.

[0094] The extracted behavioral features are formed into a grayscale behavioral image (block **210**). This may be done, for example, by taking a behavior vector including each of the respective features in a defined location in the behavior vector and form it into a two dimensional image (e.g., a 32×32 image where the vector includes 1024 or fewer features).

[0095] A first grayscale attack context image and a second grayscale attack context image are accessed from a storage medium (block **212**). The first grayscale behavioral context image and the second grayscale behavioral context image are images generated from data of prior time periods for the same trusted individual. As an example, in some embodiments, the first grayscale behavioral context image and a second grayscale behavioral context image are the preceding days' image and the image from two days prior, respectively, for the same insider being monitored. As another example, the first grayscale attack context image and a second grayscale behavioral context image are the preceding days' data and an average of the data for the preceding month, respectively, for the same insider being monitored. Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a variety of context images that may be used in relation to different embodiments.

[0096] The grayscale behavioral image is combined with the first grayscale behavioral context image and a second grayscale behavioral context image to yield a color behavioral image (block **214**). This may be done, for example, by assigning a respective component color (e.g., red, green, or blue) to each of the grayscale behavioral image, the first grayscale attack behavioral image, and the second grayscale behavioral context image. As a particular example, the values of the grayscale behavioral image are assigned as red values, the values of the first grayscale behavioral context image are assigned as green values, and the second grayscale behavioral context image are assigned as blue values. When combined, each location of the resulting image has a complemented of red, green, and blue values which render the color image.

[0097] An insider attack detection and classification model is applied to the color behavioral image (block **216**). This application results in an indication of whether the color behavioral image indicates an insider attack. This model is trained using actual insider attack images and synthetic insider attack images as discussed below, and the more images available for training generally result in an increased accuracy of the model.

[0098] Where an insider attack is indicated based upon the color behavioral image (block **218**), an expert is alerted to review the data available on the detected insider attack (block **220**). Where the expert reviews the data and indicates that it was in fact an insider attack (block **222**), the color behavioral image is stored to a storage medium as an actual insider attack image (block **224**) and the insider attack is handled by human resources (block **226**). Alternatively, where either the model indicated that the color behavioral image was not an insider attack (block **220**) or the expert indicated that it was not an inside attack (block **222**), the color behavioral image is stored as a non-insider attack image (block **228**). This non-insider attack image may be used, for example, to generate grayscale behavioral context images.

[0099] Turning to FIG. 2B, a flow diagram **250** continues the method of FIG. 2A. Following flow diagram **250**, the actual insider attack data maintained in the storage medium is used to perform insider attack multiplication (block **252**). Such insider attack multiplication results in one or more synthetic insider attack images being generated for each actual insider attack image in the storage

medium. In some embodiments, as discussed in relation to FIG. 3 below, one synthetic insider attack image is generated for each available actual insider attack image. In various embodiments, as discussed below in relation to FIG. 5, any number of synthetic insider attack images is generated for each actual insider attack image in the storage medium limited only by the number of replacement context images available and the possibility of significant overfitting. The generated synthetic insider attack images are stored with the actual insider attack images.

[0100] Both the actual insider attack images and synthetic insider attack images are processed to reduce any artefacts of data multiplication (block 254). This processing may be done similar to that discussed below in relation to FIGS. 7-9.

[0101] Using the processed insider attack images, an insider attack detection and classification model is trained (block 256). Such training may be done in accordance with any known machine learning approach known. As more insider attack images become available, the training of the insider attack detection and classification model results in more accurate understanding of what an insider attack looks like, and thus provides more accuracy in detecting and classifying insider attacks.

[0102] Turning to FIG. 3, a flow diagram 300 shows a method in accordance with some embodiments for generating one additional set of insider attack data (i.e., a synthetic insider attack image) for every actual set of insider attack data available. One of a number of actual insider attack images is selected and accessed from a storage medium (block 304).

[0103] The attack images may include the actual attack data as one channel of the images, and include context data as other channels of the data. In one embodiment, the channel of the actual insider attack data is red, and two context channels are respectively green and blue. By combining the three channels, a three color image (red, green, blue) is created as an actual insider attack image. The context channels include data for the same individual represented by the actual insider attack data for one or more periods of time where no insider attack occurred. As such, the context channels are in contrast to the channel with the actual insider attack data as, unlike the actual insider attack channel, they do not show an insider attack. As an example, in some embodiments, the two context data images are the preceding days' data and the data from two days prior, respectively, for the same individual reflected in the actual insider attack data. As another example, the two context data images are the preceding days' data and an average of the data for the preceding month, respectively, for the same individual reflected in the actual insider attack data.

[0104] The actual insider attack images have been previously classified as an insider attack. In some cases, the classification is an automatic classification using an insider attack classification model. In other cases, the classification is a manual classification made by a system administrator. In yet other cases, the classification is at first an automatic classification followed by a manual classification. The actual insider attack images are stored to a computer readable medium.

[0105] The selected actual insider attack image is disaggregated into component channels (block 306). In particular, the actual insider attack image is disaggregated into: (1) a grayscale actual attack image from a first data in the actual insider attack image, (2) a first grayscale attack context image from a second data in the actual insider attack image, and (3) a second grayscale attack context image from a third data in the actual insider attack image. The first data, second data, and third data of the actual insider attack image are the respective channels discussed above. Thus, using the example above where the three channels are red, green, and blue, respectively, the actual grayscale actual attack image may be formed from the red components of the actual insider attack image; the first grayscale attack context image may be formed from the green components of the actual insider attack image; and the second grayscale attack context image may be formed from the blue components of the actual insider attack image.

[0106] Turning to FIG. 4, a process 400 is shown where a color actual insider attack image 410 is disaggregated into a three channel set 420 with: (1) a grayscale actual attack image 422 corresponding to one component color (e.g., red) of actual insider attack image 410, (2) a first

grayscale attack context image **424** corresponding to another component color (e.g., green) of actual insider attack image **410**; and (3) a second grayscale attack context image **426** corresponding to yet another component color (e.g., blue) of actual insider attack image **410**.

[0107] Returning to FIG. **3**, the first grayscale attack context image is swapped with the second grayscale attack context image by assigning the first grayscale attack context image as the third data, and assigning the second grayscale attack context image as the second data (block **308**). The first data corresponding to actual attack image remains unchanged.

[0108] Turning to FIG. **4**, process **400** shows the swapping of first grayscale attack context image **424** with second grayscale attack context image **426** in a three channel set **430**. In the case where grayscale actual attack image **422** corresponds to the red components of actual insider attack image **410**, first grayscale attack context image **424** correspond to the green components of actual insider attack image **410**, and second grayscale attack context image **426** correspond to the blue components of actual insider attack image **410**, the value of the blue components become the value of green components, and the value of the green components become the value of blue components.

[0109] Returning to FIG. **3**, the swapped first grayscale attack context image and second grayscale attack context image are re-combined with the grayscale actual attack image to yield a synthetic insider attack image (block **310**). The color of the synthetic insider attack image will be different (unless the two swapped attack context images were identical) from the actual insider attack image. This is because two color components (e.g., red and green) will be swapped from the original. This synthetic insider attack image is stored with other insider attack images (both actual and synthetic) (block **312**). These combined insider attack images can be used to train an insider attack detection model.

[0110] Turning to FIG. **4**, process **400** shows the combining of swapped grayscale attack context images **424**, **426** with grayscale actual attack image **422** to yield a synthetic insider attack image **440**. Again, as described in relation to FIG. **3**, the color of synthetic insider attack image **440** will be different from the color of actual insider attack image **410** unless first grayscale attack context image **424** is identical to second grayscale attack context image **426**.

[0111] Returning to FIG. **3**, it is determined whether another actual insider attack image remains to be processed (block **314**). Where another remains to be processed (block **314**), the processes of blocks **304-212** are repeated for the next selected actual insider attack image.

[0112] Turning to FIG. **5**, a flow diagram **500** shows a method in accordance with some embodiments for generating multiple additional sets of insider attack data for every actual set of insider attack data available. Following flow diagram **500**, one of a number of actual insider attack images is selected and accessed from the storage medium (block **504**). The attack images may include the actual attack data as one channel of the images, and include context data as other channels of the data. In one embodiment, the channel of the actual insider attack data is red, and two context channels are respectively green and blue. By combining the three channels, a three color image (red, green, blue) is created as an actual insider attack image. The context channels include data for the same individual represented by the actual insider attack data for one or more periods of time where no insider attack occurred. As such, the context channels are in contrast to the channel with the actual insider attack data as, unlike the actual insider attack channel, they do not show an insider attack. As an example, in some embodiments, the two context data images are the preceding days' data and the data from two days prior, respectively, for the same individual reflected in the actual insider attack data. As another example, the two context data images are the preceding days' data and an average of the data for the preceding month, respectively, for the same individual reflected in the actual insider attack data.

[0113] The actual insider attack images have been previously classified as an insider attack. In some cases, the classification is an automatic classification using an insider attack classification model. In other cases, the classification is a manual classification made by a system administrator.

In yet other cases, the classification is at first an automatic classification followed by a manual classification. The actual insider attack images are stored to a computer readable medium.

[0114] In addition, replacement context images are accessed (block **506**). The replacement context images include a first replacement context image and a second replacement context image. The replacement context images are context images that are related to the same individual to which the actual insider attack image is related, but are for different time periods than the context included in the actual insider attack image. As an example, the context data included with the actual insider attack image may be the preceding days' data and the data from two days prior, respectively. In contrast, the replacement context images may be an average of the prior week and an average for the prior month for the individual to which the actual insider attack image is related. Similar to the context data included with the actual insider attack image, the replacement context images do not correspond to an insider attack, but are not the same as the context data included with the actual insider attack image.

[0115] The selected actual insider attack image is disaggregated into component channels (block **508**). In particular, the actual insider attack image is disaggregated into: (1) a grayscale actual attack image from a first data in the actual insider attack image, (2) a first grayscale attack context image from a second data in the actual insider attack image, and (3) a second grayscale attack context image from a third data in the actual insider attack image. The first data, second data, and third data of the actual insider attack image are the respective channels discussed above. Thus, using the example above where the three channels are read, green, and blue, respectively, the actual grayscale actual attack image may be formed from the red components of the actual insider attack image; the first grayscale attack context image may be formed from the green components of the actual insider attack image; and the second grayscale attack context image may be formed from the blue components of the actual insider attack image.

[0116] Turning to FIG. **6**, a process **600** is shown where an image set **610** including a color actual insider attack image **612** is accessed along with a first replacement context image **614** and a second replacement context image **616**. An image set **620** shows a color actual insider attack image **612** that is disaggregated into: (1) a grayscale actual attack image **622** corresponding to one component color (e.g., red) of actual insider attack image **612**, (2) a first grayscale attack context image **624** corresponding to another component color (e.g., green) of actual insider attack image **612**; and (3) a second grayscale attack context image **626** corresponding to yet another component color (e.g., blue) of actual insider attack image **612**.

[0117] Returning to FIG. **5**, the first grayscale attack context image from the second data in the actual insider attack image is replaced by the first replacement context image, and the second grayscale attack context image from the third data in the actual insider attack image is replaced by the second replacement context image (block **510**). This is done by assigning the first replacement context image as the second data, and assigning the second replacement context image as the third data.

[0118] Turning to FIG. **6**, process **600** shows replacement of first grayscale attack context image **624** with first replacement context image **614**, and replacement of second grayscale attack context image **626** with second replacement context image **616** to yield an image set **630**.

[0119] Returning to FIG. **5**, the replacement context images are combined with the grayscale actual attack image to yield a synthetic insider attack image (block **512**). The color of the synthetic insider attack image will be different (unless the two replacement context images are identical to those they are replacing) from the actual insider attack image. This synthetic insider attack image is stored with other insider attack images (both actual and synthetic) (block **514**). These combined insider attack images can be used to train an insider attack detection model.

[0120] Turning to FIG. **6**, process **600** shows the combining of replacement context images **614**, **616** with grayscale actual attack image **622** to yield a synthetic insider attack image **640**. A gain, as described in relation to FIG. **5**, the color of synthetic insider attack image **640** will be different

from the color of actual insider attack image **612** unless replacement context image **614** is identical to first grayscale attack context image **624** and replacement context image **616** is identical to first grayscale attack context image **626**.

[0121] Returning to FIG. **5**, it is determined whether more replacement context images remain to replace grayscale context images from the actual attack vector image (block **516**). Any number of replacement context images can in theory be used, with each set of replacement context images resulting in generation of another synthetic insider attack image. However, at some number of replacement context images the resulting synthetic insider attack images start to be impacted significantly by overfitting. Where more replacement context images remain (block **516**), the processes of blocks **506-416** are repeated for the next set of replacement context images.

[0122] Where no additional replacement context images remain (block **516**), it is determined whether another actual insider attack image remains to be processed (block **518**). Where another remains to be processed (block **516**), the processes of blocks **504-418** are repeated for the next selected actual insider attack image.

[0123] Turning to FIG. **7**, a novel GANs architecture **700** tailored for revealing insider attacks is shown in accordance with various embodiments. As shown, novel GANs architecture **700** includes a generator **740** and a critic **750**. Generator **740** is configured to generate an image for each vector it receives. Where known insider attacks are available, such vectors corresponding to the insider attacks are fed into generator **740** and each result in a respective image. In addition, generator **740** introduces noise to the received insider attack vectors by making small modifications to different vector values and generates images for each of these noise modified vectors. Such an approach results in a significant multiplication of images for the actual insider attack vectors and for surrounding noise modified vectors. Such an approach yields potentially thousands of slightly different images for each of the actual insider attack vectors.

[0124] Critic **750** is configured to accept a generated image from generator **740** and classifies the received image as a real image or a generated image. In effect, critic **750** seeks to determine whether an image received from generator **740** was created using a vector from an actual insider attack or a noise modified vector. Where the noise modification was to vector elements that are not indicative of an insider attack, critic **750** is able to classify the received image as not corresponding to an insider attack or a generated image. In contrast, where the noise modification was to vector elements that are indicative of an insider attack, critic **750** is less able to correctly classify the received image as a generated image. This classification information is fed back to generator **740**.

[0125] In operation, generator **740** begins by adding noise to random elements in a known insider attack vectors to generate a large number of images. These images are provided to critic **750** that operates to classify the received images as either real images (i.e., generated from an actual insider attack) or generated images (i.e., generated from a noise modified vector). This classification information is fed back to generator **740**. Where a noise modified vector is incorrectly classified by critic **750** as a real image, generator **740** learns that modification around the vector element modified in the particular noise modified vector confuses critic **750**. In contrast, where a noise modified vector is correctly classified by critic **750** as a generated image, generator **740** learns that modification around the vector element modified in the particular noise modified vector does not confuse critic **750**.

[0126] The desired result is for critic **750** to show an inability to properly classify or to return a correct classification half of the time and an incorrect classification half of the time. The greater the confusion of critic **750**, the higher likelihood that generator **740** has identified elements within a network activity vector that are relevant to discerning and properly classifying network activity as an insider attack. Thus, over time as generator **740** learns which elements of a network activity vector when modified causes misclassification by critic **750**, generator focuses more and more on the identified elements as it learns which elements of a network activity vector are relevant to identifying an insider attack. Once a threshold misclassification by critic **750** is achieved, the model

is considered trained and may be deployed for identifying insider attacks in generic network activity ongoing in a network. In some embodiments, a model is considered trained when the misclassification is greater than forty percent (40%). In other embodiments, a model is considered trained when the misclassification is greater than forty-five percent (45%). In yet other embodiments, a model is considered trained when the misclassification is greater than forty-nine percent (49%). Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a variety of thresholds that may be used to define a model as properly trained.

[0127] As shown, generator **740** includes a latent image dimension stage **702** where a size of the image is defined. In some embodiments, the image is defined as including one hundred, seventy (170) vector elements. Based upon the disclosure provided herein, one of ordinary skill art will recognize a variety of sizes that may be used in relation to different embodiments. The vector of the defined size is provided to a leaky rectified linear unit (leaky ReLU) fully connected layer stage **704**. Next, normalization is performed in a batch normalization stage **706**. A second leaky ReLU fully connected layer stage **708** is applied to the normalized output, and the result is again normalized in a batch normalization stage **710**.

[0128] The combination of the leaky rectified linear unit (leaky ReLU) fully connected layer stages **704**, **708** are used to generate specificity information. An example of specificity is shown in FIG. **9A** where black dots **911**, **921**, **931**, **941** represent particular data are maintained in the same location across multiple images **910**, **920**, **930**, **940**. This avoids the situation where the model becomes trained based upon the movement of particular data across images.

[0129] An image reshaping stage **712** is performed to render the image of a defined shape. In some embodiments, the image shapes is a 256 bit gray scale 4×4 image. A leaky ReLU two dimension convolution stage **714** is performed followed by normalization in a batch normalization stage **716**. In some embodiments, leaky ReLU two dimension convolution stage **714** the parameter f of the convolution is sixty-four (64) and the parameter k of the convolution is five (5). Another leaky ReLU two dimension convolution stage **718** is performed followed by normalization in a batch normalization stage **720**. In some embodiments, leaky ReLU two dimension convolution stage **718** the parameter f of the convolution is sixty-four (64) and the parameter k of the convolution is nine (9). Another leaky ReLU two dimension convolution stage **722** is performed followed by flattening in a flattening stage **724** and normalization in a batch normalization stage **726**. In some embodiments, leaky ReLU two dimension convolution stage **722** the parameter f of the convolution is three (3) and the parameter k of the convolution is seventeen (17).

[0130] The three convolution stages **714**, **718**, **722** are used to generate spatial information. In embodiments discussed herein, the spatial information is color information. By applying the convolution stages **714**, **718**, **722** a generated image will have the same color across the image. An example of such spatial data uniformity is shown in FIG. **9A** where image **910** is uniformly a first color, image **920** is uniformly a second color, image **930** is uniformly a third color, and image **940** is uniformly a fourth color. In contrast the example in FIG. **9B** was generated without all of the convolution stages. An image **960** of FIG. **9B** corresponds to image **910** of FIG. **9A**; [0131] an image **970** of FIG. **9B** corresponds to image **920** of FIG. **9A**; [0132] an image **980** of FIG. **9B** corresponds to image **930** of FIG. **9A**; and an image **990** of FIG. **9B** corresponds to image **940** of FIG. **9A**. Again, image **910** has a uniform first color across the image. In contrast, image **960** has a non-uniform color with a diagonal center section **961** being of one color and the outer corner areas **962**, **963** being another color.

[0133] Returning to FIG. **7**, a Tan h fully connected layer stage **728** is applied followed by an image reshape stage **730**. In some embodiments, Tan h fully connected layer stage **728** is done with a parameter of three thousand, seventy-two (3072), and the image is reshaped to a three color (i.e., RGB) 32×32 image.

[0134] Critic **750** receives an image generated by generator **740** in an image input stage **752**. Critic **750** flattens the received input image in a flattening stage **754**. Critic applies a leaky ReLU) fully

connected layer stage **756** with a parameter of three thousand, seventy-two (3072) and performs an image reshape stage **758** where the image is reshaped into a three color (i.e., RGB) 32×32 image. [0135] Critic applies three successive leaky ReLU two dimension convolution stages **760**, **762**, **764** are applied. For the leaky ReLU two dimension convolution stage **760**, the parameter f of the convolution is sixty-four (64) and the parameter k of the convolution is seventeen (17). For the leaky ReLU two dimension convolution stage **762**, the parameter f of the convolution is one-hundred, twenty-eight (128) and the parameter k of the convolution is nine (9). For the leaky ReLU two dimension convolution stage **764**, the parameter f of the convolution is two-hundred, fifty-six (256) and the parameter k of the convolution is five (5). The result is flattened in a flattening stage **766**.

[0136] Another leaky ReLU fully connected layer stage **768** is applied to the flattened result, and sigmoid output layer stage **770** is applied to the result. The output of sigmoid output layer stage **770** is a confidence value indicating whether the image processed corresponds to a real image or a generated image. This classification information is passed back to generator **740** where it is used to influence which elements of received vectors are relevant to identifying an insider threat.

[0137] Turning to FIG. **8**, a flow diagram **800** shows a method of revealing insider attacks in accordance with some embodiments. Following flow diagram **800**, a vectorized data set is received (block **802**). The vectorized data set includes data from network traffic that were identified as insider attacks. As the number of insider attacks that occur is a very small percentage of network traffic, only a relatively small number of vectorized data sets are available for insider attacks. Various data or characteristics associated with the network traffic are assembled into a vector to yield a network activity vector, and multiple of the network activity vectors make up the vectorized data set. The data or characteristics of a network activity may include, but are not limited to, an indication of the source of the network activity, an indication of the target of the network activity, an indication of whether the source of the network activity is an insider, an indication of whether the source of the activity is an outsider, an indication of the action requested by the network activity, an indication of a time of the network activity, an indication of the use of a thumb drive or other mobile storage, and the like. Based upon the disclosure provided herein, one of ordinary skill in the art will recognize a large number of data or characteristics of a network activity that may be formed into a vectorized data set in accordance with different embodiments.

[0138] A variety of convolution layers and fully connected layers formed to operate as a generator are applied to the vectorized data set (block **804**). In some embodiments, the generator is a novel GANs generator. In a particular embodiment, the GANs generator includes a latent image dimension stage where a size of the image is defined. In some embodiments, the image is defined as including one hundred, seventy (170) vector elements. A leaky ReLU fully connected layer stage is applied to the received vectors, and normalization is performed in a batch normalization stage. A second leaky ReLU fully connected layer stage is applied to the normalized output, and the result is again normalized in a batch normalization stage.

[0139] An image reshaping stage is performed to render the image of a defined shape. A leaky ReLU two dimension convolution stage is performed followed by normalization in a batch normalization stage. Another leaky ReLU two dimension convolution stage is performed followed by normalization in a batch normalization stage. Another leaky ReLU two dimension convolution stage is performed followed by flattening in a flattening stage and normalization in a batch normalization stage. A Tan h fully connected layer stage is applied followed by an image reshape stage. This yields an image corresponding to the received vector.

[0140] The images generated for each vector are provided to a critic that in turn applies a combination of convolution layers and fully connected layers in an attempt to classify the respective image as an image generated from actual insider attack data (i.e., a real image) or an image generated based upon a modified version of actual insider attack data (i.e., a generated image) (block **806**). In some embodiments, the critic flattens the image received from the generator

and applies a leaky ReLU fully connected layer stage with a parameter of three thousand, seventy-two (3072) and performs an image reshape stage where the image is reshaped into a three color (i.e., RGB) 82×32 image. The critic then applies three successive leaky ReLU two dimension convolution stages. Another leaky ReLU fully connected layer stage is applied to the flattened result, and sigmoid output layer stage is applied to the result. The output of sigmoid output layer stage is a confidence value indicating whether the image processed corresponds to a real image or a generated image.

[0141] After an initial number of images have been generated by the generator and classified by the critic (block **807**), it is determined whether classification accuracy of the critic has gone below a defined threshold (block **808**). Where the classification accuracy by the critic is not below the threshold (block **807**), additional vectors are generated by adding noise to different elements of the actual insider attack data (block **810**). Over time the generator begins to understand to which elements noise can be added and yet the critic will identify an image generated based upon the noise modified vector as having been generated based upon actual insider attack data, and which elements when noise modified result in correct identification as having been generated from noise modified data. Thus, each time the system loops more emphasis is made on generating noise modified vectors where the noise is introduced to the elements where it is more likely to fool the critic. The processes of blocks **804-306** are repeated until the accuracy of the critic classification goes below the defined threshold (block **808**).

[0142] Once the accuracy of the critic classification goes below the defined threshold (block **808**), the model is considered trained (block **812**) and is deployed to detect insider attacks based upon received network traffic data (block **814**).

[0143] Also described herein is a technique that uses transfer learning in concert with non-dynamic data to improve classification model accuracy for identifying malicious behavior. In some embodiments, transfer learning with ResNet50 is used for image encoding. A dual-input classifier is provided that processes image-encoded behavioral data in concert with the non-dynamic data, such as user role and psychometric information. In some embodiments, transfer learning is applied using the ResNet50 architecture, although other model architectures may also be used in those or other embodiments. After training ResNet50 on the ImageNet dataset, the entire architecture is frozen except for the last residual block, allowing for the final features extracted from the model to be tweaked during training.

[0144] While the image encoding enables the identification of malicious behavior at a glance, it is beneficial to provide additional information regarding a user, such as the user's role in the organization, to the classification model. For example, keylogger attacks are more likely to be performed by users whose role relates to information technology (IT) services within and organization, as those users have the requisite access to execute such an attack. Additionally, logging into a computer after hours is not a sign of malicious behavior for a security guard that is on a night shift. Accordingly, use role is a useful factor in determining whether observed behavior is malicious and for determining the potential types of insider attacks that are available to a user.

[0145] In some embodiments, psychometric information is incorporated in addition to role information. This psychometric information provides the model information into a person's psyche and assists in determining if that person may be more likely to perform insider attacks. While most employees do not sit with psychologists on the job where such information can be easily compiled, it is becoming more common over time for companies to require psychometric tests to be taken by potential candidates or incorporate psychometric properties into the hiring process, for example, scoring metrics such as the Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism (OCEAN) model.

[0146] Insider threat researchers have also identified several other non-dynamic factors that indicate that insider may be predisposed to becoming an insider threat, such as but not limited to previous policy violations or disciplinary actions. Additionally, a previous or recent demotion in

role or rank within an organization may be a factor.

[0147] In some embodiments, feature vectors for these non-dynamic features are concatenated with the output of the ResNet50 architecture which is a 50-layer deep residual convolutional neural network. The concatenation allows the non-dynamic data to be integrated into the output before the combined data is passed to a set of fully connected layers for classification. Although the ResNet50 architecture is used in some embodiments, other embodiments make use of other architectures, such as but not limited to variants of the MobileNet architecture or the VGG19 architecture.

[0148] Turning to FIG. 10, an attack classification architecture **1000** is shown in accordance with some embodiments. The attack classification architecture **1000** provides a dual input classifier that classifies user behavior based on a behavioral image **1001** and non-dynamic information **1002** to output a classification of encoded behavior as benign or malicious. The behavioral image **1001** is generated based on behavioral encoding techniques described herein. The non-dynamic information **1002** includes but is not limited to the user's role within an organization, known psychometric data (e.g., OCEAN), or other non-dynamic factors associated with the user. The behavioral image **1001** may be a color behavioral image as described herein that encodes user behavior for a given time period, along with behavioral context for preceding time periods, within different color channels. While in some embodiments, the behavioral image **1001** may be sufficient to classify a given set of encoded user behavior as benign or malicious based on the encoded behavior and the behavioral context encoded into the behavioral image **1001**, the concatenation of non-dynamic information can enhance the accuracy of the classification produced by the attack classification architecture **1000**.

[0149] In some embodiments, the attack classification architecture **1000** processes the behavioral image **1001** via a ResNet50 architecture **1004** that has been trained via transfer learning using the training dataset described above. Output of the ResNet50 architecture **1004** is then be provided to fine-tuning layers including an average pooling layer **1006** and a series of fully connected layer stages that are coupled via batch normalization layers. Before input is provided to the fully connected layer stages, a concatenation layer **1008** concatenates the output of the average pooling layer **1006** with the non-dynamic information **1002** that is available for the user. The output of the concatenation layer **1008** is provided to the series of fully connected layer stages with associated batch normalization layers. The fully connected layers use a ReLU activation function and have an L2 regularization term that adds a penalty proportional to the squared Euclidean (L2) norm of the layer's weight matrix to discourage excessively large weight values. A first fully connected layer stage **1010** receives output from the concatenation layer **1008** as input and processes the input. The first batch normalization layer **1012** normalizes the output of the first fully connected layer stage **1010** before that output is provided as input to a second fully connected layer stage **1014** and associated second batch normalization layer **1016**. A third fully connected layer stage **1018** and associated third batch normalization layer **1020** then process the output from the second batch normalization layer **1016**. The output of the third batch normalization layer **1020** is then processed by a fourth fully connected layer **1022** having a Sigmoid activation function. In some embodiments, the first fully connected layer stage **1010** has 1024 hidden layers, the second fully connected layer stage **1014** has 512 hidden layers, and the third fully connected layer state **1018** has 256 hidden layers, although other layer structures may also be used. The fourth fully connected layer **1022** provides a classification output that classifies the behavior encoded behavioral image **1001** and non-dynamic information **1002**. The classification performed based on the behavioral image **1001** and non-dynamic information **1002** has been determined to be more accurate than classifications performed based solely on the behavioral image **1001**.

[0150] The flow diagram **200** of a method of detecting insider attacks and training an insider attack detection and classification model shown in FIG. 2A-2B can be updated to the flow diagram **1100** shown in FIG. 11A-11B. As shown in FIG. 11A, the flow diagram **1100** for the updated method begins similarly to the flow diagram **200** of FIG. 2A-2B. However, after the generation of the color

behavioral image (block **214**), the attack classification architecture **1000** selects non-dynamic information regarding the trusted insider (block **1102**). The selected non-dynamic information may include some or all of the non-dynamic information available for the trusted user based on tuning or configuration parameters for the attack classification architecture **1000**. For example, in some embodiments the user role is included by default within the non-dynamic information, while other psychometric information such as OCEAN metrics, disciplinary records, or data indicating a recent demotion or other role change within the organization. As shown in the flow diagram **1100** of the updated method, the attack classification architecture **1000** can then pass color behavior image and non-dynamic information into a dual input classifier of the attack classification architecture **1000** (block **1104**), where the dual input classifier has been trained based on training data that includes synthetic insider attack images as described herein, in conjunction with non-dynamic training data. The attack classification architecture **1000** can then apply the trained insider attack detection and dual input classification model to the color behavior image and the non-dynamic information (block **1106**) to determine a classification score for the input, which indicates the likelihood that the input represents an insider attack.

[0151] Example signs of a potential insider attack may be, for example, a user that begins accessing sensitive or confidential information while also having a pattern of visiting websites that are known to be used to leak confidential information. Additionally, if a user with a pattern of browsing job sites begins accessing confidential information that is outside of the user's normal access pattern, the user may intend to steal that confidential information before leaving for a different job. In one scenario, a system administrator that is known to be disgruntled may be detected as having downloaded a keylogger for installation onto their supervisor's or co-worker's computer to acquire their password or other information. Additionally, insiders with IT roles may be detected to have performed activities such as reconfiguring firewalls, installing malware, or facilitating unauthorized external access to an organization's network.

[0152] Turning to FIG. **11B**, the dual input classifier of the attack classification architecture **1000** can classify the input as indicative of an insider attack or a non-insider attack (block **218**). The classifier can indicate that the input indicates an insider attack when a classification score exceeds a threshold value. Where an insider attack is indicated based upon the color behavioral image and non-dynamic data (block **218**), an expert is alerted to review the data available on the detected insider attack (block **220**). Where the expert reviews the data and indicates that it was in fact an insider attack (block **222**), the color behavioral image and non-dynamic information pairing is stored to a storage medium as actual insider attack data (block **1110**) and the insider attack is handled by human resources (block **226**). Alternatively, in response to the dual input classifier indicating that the color behavioral image and non-dynamic data pair do not represent an insider attack (block **220**) or the expert indicating that the color behavioral image and non-dynamic data pair was not an insider attack (block **222**), the color behavioral image and non-dynamic information pairing is stored as a non-insider attack data (block **1108**), or non-attack data in general, such that the classified behavior was determined not to be malicious in the context of general user behavior or when considered in the context of the non-dynamic data.

[0153] Whether the input is determined to indicate actual insider attack data or non-insider attack (e.g., benign) data, the data can be used to update insider attack dual input classifier model training data (block **1112**). In some embodiments, the pairing of color behavioral image and non-dynamic data can be stored with the classification score output by the dual input classifier. In some embodiments, the pairing is stored with a binary or Boolean value that indicates whether the classification score was over the insider attack classification threshold. In such embodiments, the insider attack classification threshold value may also be stored. The dual input classifier can then be retrained with the updated training data (block **1114**). The attack classification architecture **1000** can then receive deployment of a re-trained dual-input classifier (block **1116**) having increased accuracy.

[0154] In conclusion, the above disclosure provides novel systems, devices, and methods for insider thread identification. While detailed descriptions of one or more embodiments of the have been given above, various alternatives, modifications, and equivalents will be apparent to those skilled in the art without varying from the general spirit of the above disclosed embodiments. Therefore, the above description should not be taken as exemplary, rather than limiting, without departing from the broader spirit and scope of the features set forth in the appended claims.

Claims

1. A method for detecting behavior indicative of an insider attack, the method comprising: logging, by a processing resource, activities associated with an electronic account of an insider within an organization, the activities performed in relation to a communication network to yield logged activities; extracting, by the processing resource, a set of defined behavioral features from the logged activities; forming, by the processing resource, color behavioral image based on a grayscale behavioral image corresponding to the set of defined behavioral features and one or more grayscale behavioral context images; and applying, by the processing resource, a multiple-input insider attack classification model to the color behavioral image and non-dynamic data associated with the insider, the non-dynamic data including an organizational role of the insider.
2. The method of claim 1, comprising: generating a classification score via the multiple-input insider attack classification model; and determining whether the classification score indicates an insider attack based on a comparison with an insider attack classification threshold.
3. The method of claim 2, comprising: storing, by the processing resource, the color behavioral image and non-dynamic data as actual insider attack data in response to determining that the classification score indicates an insider attack; and storing, by the processing resource, the color behavioral image and non-dynamic data as non-attack data in response to determining that the classification score does not indicate an insider attack.
4. The method of claim 3, comprising: re-training the multiple-input insider attack classification model based at least in part on the actual insider attack data and the non-attack data; and deploying the multiple-input insider attack classification model after re-training.
5. The method of claim 1, wherein the non-dynamic data includes psychometric data associated with the insider.
6. The method of claim 1, wherein the non-dynamic data includes disciplinary record data associated with the insider.
7. The method of claim 1, wherein the one or more grayscale behavioral context images include an encoding of activities associated with the electronic account that were logged during a different time period relative to the logged activities.
8. The method of claim 7, wherein the one or more grayscale behavioral context images include an encoding of activities associated with the electronic account that were logged during a previous time period relative to the logged activities.
9. A non-transitory computer-readable medium having stored therein instructions that when executed by a processing resource cause the processing resource to perform operations comprising: logging activities associated with an electronic account of an insider within an organization, the activities performed in relation to a communication network to yield logged activities; extracting a set of defined behavioral features from the logged activities; forming color behavioral image based on a grayscale behavioral image corresponding to the set of defined behavioral features and one or more grayscale behavioral context images; and applying a multiple-input insider attack classification model to the color behavioral image and non-dynamic data associated with the insider, the non-dynamic data including an organizational role of the insider.
10. The non-transitory computer-readable medium of claim 9, wherein logging the activities associated with an electronic account includes logging a timestamp associated with the activities.

- 11.** The non-transitory computer-readable medium of claim 10, wherein the activities associated with the electronic account of the insider within an organization includes one or more of: file systems accessed, files accessed, computers used, applications used, and use of external storage media.
 - 12.** The non-transitory computer-readable medium of claim 9, wherein applying the multiple-input insider attack classification model to the color behavioral image and non-dynamic data includes: processing the color behavioral image via a first neural network model; concatenating output of the first neural network model with the non-dynamic data to generate concatenated data; processing the concatenated data via a second neural network model to generate a classification score; and determining whether the classification score indicates an insider attack based on a comparison with an insider attack classification threshold.
 - 13.** The non-transitory computer-readable medium of claim 12, wherein the first neural network model includes a residual convolutional neural network trained via transfer learning.
 - 14.** The non-transitory computer-readable medium of claim 12, wherein the second neural network model includes a plurality of fully connected layers that are interconnected via a plurality of batch normalization layers.
 - 15.** The non-transitory computer-readable medium of claim 14, wherein at least a portion of the plurality of fully connected layers include a L2 regularization term.
 - 16.** An apparatus to determine a classification for insider behavior, the apparatus comprising: a memory device configured to store instructions; and a processing resource configurable to execute the instructions, wherein the instructions configure the processing resource to: generate a color behavioral image based on a grayscale behavioral image corresponding to a set of defined behavioral features logged for an electronic account associated with an insider of an organization and one or more grayscale behavioral context images; process the color behavioral image via a first neural network model; concatenate output of the first neural network model with non-dynamic data associated with the insider to generate concatenated data; process the concatenated data via a second neural network model to generate a classification score; and determine whether the classification score indicates an insider attack based on a comparison with an insider attack classification threshold.
 - 17.** The apparatus of claim 16, wherein the processing resource is configured select the non-dynamic data associated with the insider from a set of non-dynamic data that includes a role of the insider within the organization, psychometric data, and a disciplinary record associated with the insider.
 - 18.** The apparatus of claim 16, wherein the processing resource is configurable to concatenate output of the first neural network model with the non-dynamic data via a concatenation layer positioned logically between the first neural network model and the second neural network model.
 - 19.** The apparatus of claim 16, wherein the first neural network model includes a residual convolutional neural network trained via transfer learning and the second neural network model includes a series of fully connected layer stages, including at least a first fully connected layer stage, a second fully connected layer stage, and a third fully connected layer stage, each of the fully connected layer stages having a ReLU activation function and an associated batch normalization layer.
 - 20.** The apparatus of claim 19, wherein the second neural network model additionally includes a fourth fully connected layer having a Sigmoid activation function, the fourth fully connected layer configurable to receive output of a batch normalization layer associated with the third fully connected layer stage and provide a classification output to classify the set of defined behavioral features as benign or malicious.
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